

COAL (Computer Organization and Assembly Language) (EE2003)

Project

Date _____

Total Marks: **50**

Total Questions: **1**

Course Instructor

Sana Fatima

Roll No

Section

Student Signature

Important Instructions: *In case of Plagiarism all assignments, projects and quizzes will be marked as zero. In case of cheating from any AI tool all quizzes will be marked zero.*

Assembly Language Game

You must pick one of the following games and implement it in **Assembly language**.
You can work individually or in a group of two.

Available Games Options:

1. Snake Game
2. Brick Breaker
3. Two-Player Pong (No AI)
4. Flappy Bird (Simplified)

Each game must meet all listed acceptance criteria and handle all edge cases.

Rules:

- No file handling, system management, or external libraries.
- Only use basic input, output, and graphics methods allowed in Assembly.
- The final submission must be fully functional and demonstrate all required features.
- Any case of plagiarism will result in a zero for all involved students.
- You must follow all mentioned requirements and edge case handling for the project to be accepted.

Note: All games must handle all edge cases.

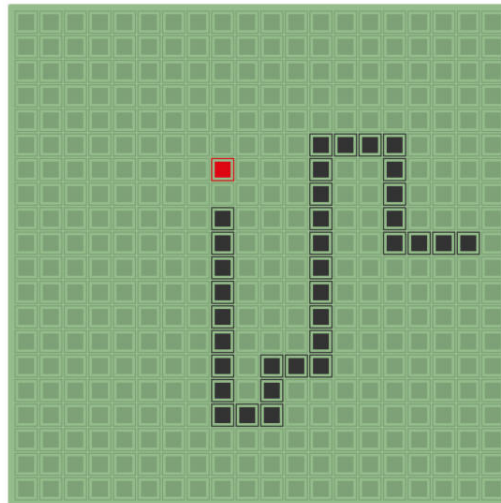
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Games Description & Acceptance Criteria

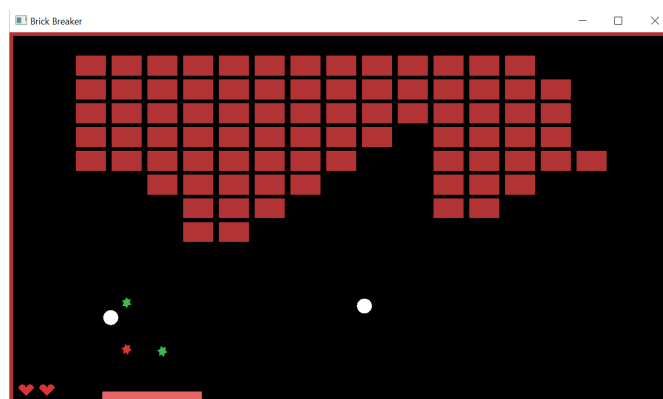
1. Snake Game

- Snake moves continuously and grows when it eats food.
- Food appears at random positions on the screen.
- Game ends when the snake hits itself or the wall.
- Score updates correctly with each food eaten.



2. Brick Breaker

- Paddle moves left and right with keyboard input.
- Ball bounces correctly off the paddle, walls, and bricks.
- Bricks disappear when hit.
- Game ends when all bricks are cleared or ball falls below paddle.
- Score increases with each brick broken.

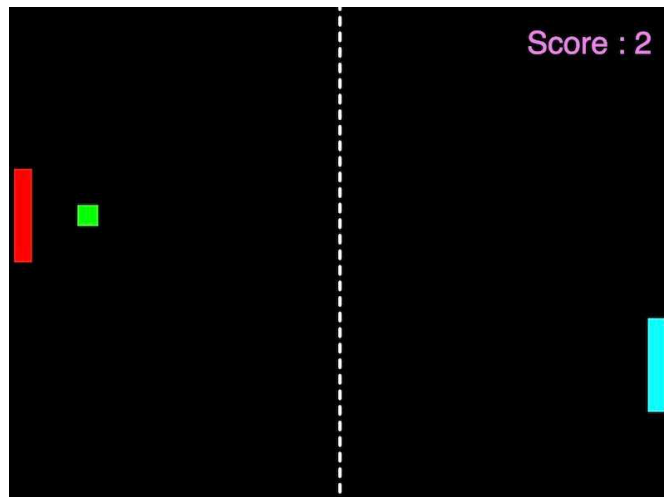


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3. Two-Player Pong (No AI)

- Two paddles controlled by different keys.
- Ball bounces off paddles and screen edges.
- Point scored when one player misses the ball.
- Score displayed and updates correctly.
- Option to restart after a win.



4. Flappy Bird (Simplified)

- Bird moves up when key is pressed and falls otherwise.
- Pipes (or obstacles) move horizontally across the screen.
- Collision with a pipe or ground ends the game.
- Score increases when the bird passes a pipe.

